

Quiz 1

601.471/671 NLP: Self-Supervised Models

Johns Hopkins University Fall 2026

Name: _____

Instructor version

Time: 15 minutes. Total: 100 points + 15 possible extra-credit points.

Complete all required questions. Q3(d) is optional extra credit.

Select one answer for each multiple-choice item. Show brief work for calculations; exact fractions and logarithms are acceptable. No calculator is needed.

Q1. Concepts and Foundations

15 points

- (a) (5 points) A training program takes the raw sentence “small birds build nests” and creates the input “small birds build” with target “nests.” What makes this example self-supervised?
- ☐ The target is a human-written label describing the sentence’s topic.
 - ☐ The target is constructed automatically from the original text.
 - ☐ The model has no prediction target.
 - ☐ The model must predict every training example correctly.

Solution: The second choice. The observed next word supplies the target without a separate annotation.

- (b) (5 points) Which equality introduces the **bigram modeling assumption**, rather than following from the probability chain rule alone?
- ☐ $P(a, b, c) = P(a)P(b | a)P(c | a, b)$
 - ☐ $P(a, b) = P(a)P(b | a)$
 - ☐ $P(c | a, b) = P(c | b)$
 - ☐ $P(a, b, c) = P(a, b)P(c | a, b)$

Solution: The third choice. It discards the earlier context a when predicting c .

- (c) (5 points) A sufficiently wide MLP with a suitable nonlinear activation can approximate a continuous target function arbitrarily well on a compact domain. What does this statement establish?
- ☐ The architecture has sufficient representational capacity.
 - ☐ Training is guaranteed to find the required parameters.
 - ☐ The fitted network is guaranteed to generalize to unseen examples.
 - ☐ Adding depth always improves test accuracy.

Solution: The first choice. Representational capacity alone guarantees neither successful optimization nor generalization.

Q2. Counting and Language-Model Evaluation

40 points

Train an **unsmoothed bigram LM** on exactly these four sentences:

```
BOS birds sing EOS
BOS birds fly EOS
BOS birds sing EOS
BOS fish swim EOS
```

Repeated sentences count separately. BOS and EOS mark sentence boundaries. Use $P(v | u) = C(u, v)/C(u)$, counting contexts within sentences.

- (a) (15 points) Compute $P(\text{birds} | \text{BOS})$, $P(\text{sing} | \text{birds})$, and $P(\text{swim} | \text{birds})$.

Solution: $3/4$, $2/3$, and $0/3 = 0$, respectively. 5 points each.

- (b) (10 points) Compute $P(\text{BOS birds sing EOS})$. Assume $P(\text{BOS}) = 1$ and include the probability of EOS.

Solution: $1 \cdot (3/4) \cdot (2/3) \cdot (2/2) = 1/2$. 5 points for the complete product (including EOS), 5 for the correct value.

- (c) (15 points) Separately, a model assigns next-token conditional probabilities $1/2$, $1/4$, and $1/8$ at the three evaluated positions of a held-out sequence. Compute the average negative log-probability H in **bits** and the perplexity 2^H .

Solution: $H = -\frac{1}{3}[\log_2(1/2) + \log_2(1/4) + \log_2(1/8)] = (1 + 2 + 3)/3 = 2$ bits per token; perplexity = 4. Scoring: log-average setup 5, H value 5, perplexity 5.

Q3. Feedforward Networks

30 + 15 bonus points

Consider this network. Both linear layers include a trainable bias:

```
hidden = nn.Linear(3, 4, bias=True)
output = nn.Linear(4, 2, bias=True)
h = torch.relu(hidden(x))
scores = output(h)
```

Each linear layer computes an affine map $Wx + b$ on the last input dimension. Like sigmoid, **ReLU (Rectified Linear Unit) is a nonlinear activation function**. It is defined elementwise by $\text{ReLU}(z) = \max(0, z)$.

- (a) (10 points) If x has shape $(5, 3)$, what are the shapes of h and $scores$?

Solution: h : $(5, 4)$; $scores$: $(5, 2)$. 5 points each.

- (b) (10 points) How many trainable parameters does the network have? Include both weights and biases, and show the count for each layer.

Solution: Hidden: $3 \cdot 4 + 4 = 16$. Output: $4 \cdot 2 + 2 = 10$. Total: 26. 5 points per correct layer count, with the total stated. ReLU has no trainable parameters.

- (c) (10 points) For one hidden neuron, let $w = (2, 1, -1)^\top$, $b = 1$, and take one input $x = (1, -2, 2)^\top$. Compute its preactivation $z = w^\top x + b$ and its ReLU activation.

Solution: $z = 2(1) + 1(-2) + (-1)(2) + 1 = -1$; $\text{ReLU}(z) = 0$. 5 points each.

- (d) **Optional bonus (15 extra-credit points).** Let (W_1, b_1) and (W_2, b_2) denote the hidden and output layers' weights and biases. Remove ReLU and write the resulting function as $W_{\text{eff}}x + b_{\text{eff}}$. Can thresholding one output represent XOR of the first two binary input coordinates, with the third fixed to zero? Explain in one sentence.

Solution: $W_{\text{eff}} = W_2 W_1$ and $b_{\text{eff}} = W_2 b_1 + b_2$ (5 points each, by expanding the composition). No: an affine threshold gives a linear boundary, but XOR is not linearly separable (5 points with reason).

Q4. One Perceptron Update

15 points

A perceptron predicts $o = \mathbf{1}[w^\top x + b > 0]$. For this exercise, update both weights and bias using

$$w' = w + \eta(t - o)x, \quad b' = b + \eta(t - o).$$

Let $w = (1, -2)^\top$, $b = 0$, $x = (1, 1)^\top$, target $t = 1$, and $\eta = 1$.

- (a) (5 points) Compute the current score $w^\top x + b$ and prediction o .

Solution: The score is $1 - 2 = -1$, so $o = 0$.

- (b) (5 points) Compute w' and b' after one update.

Solution: $w' = (1, -2)^\top + (1, 1)^\top = (2, -1)^\top$ and $b' = 1$.

- (c) (5 points) Compute the new score on the same input and its new prediction.

Solution: The score is $2 - 1 + 1 = 2$, so the new prediction is 1.